

Discrete Mathematics

Graph Theory



Types of Graphs

Null Graph

A graph which contains only an isolated vertex is called a null graph.

That is, the set of edges in a null graph is empty. Null graph with n vertices is denoted by N_n .

Example

v_1 •

• v_4

v_2 •

• v_3

Note that each vertex of a null graph is isolated.

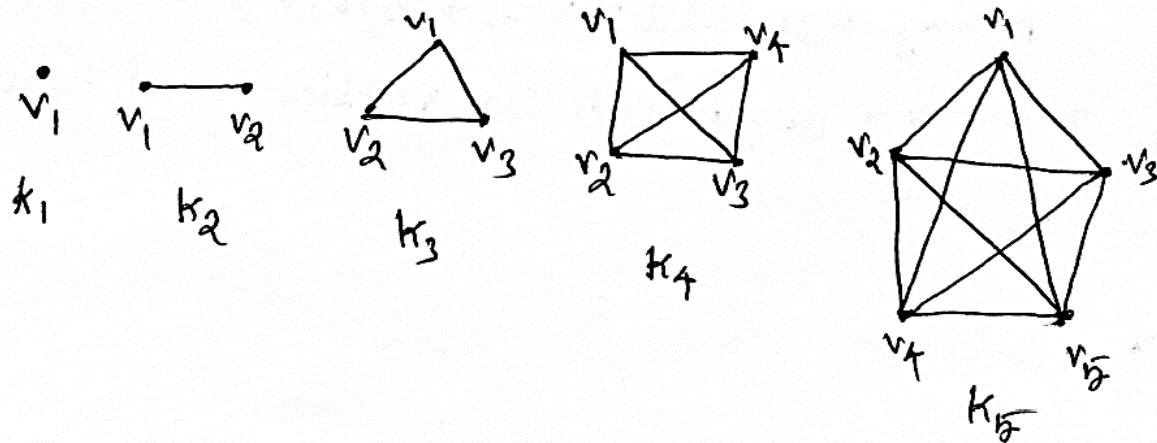
Complete Graphs

A simple graph G is said to be complete if every vertex in G is connected with every other vertex

ie, If G contains exactly one edge between each pair of distinct vertices.

A complete graph is usually denoted by K_n

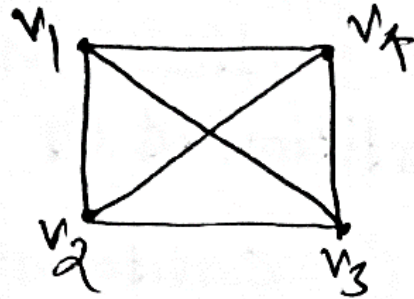
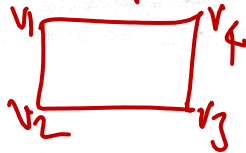
Complete graphs K_n for the vertices 1, 2, 3, 4, 5



Regular Graphs

A graph in which all vertices are of equal degree is called a regular graph. If the degree at each vertex is r , then the graph is called a regular graph of degree r .

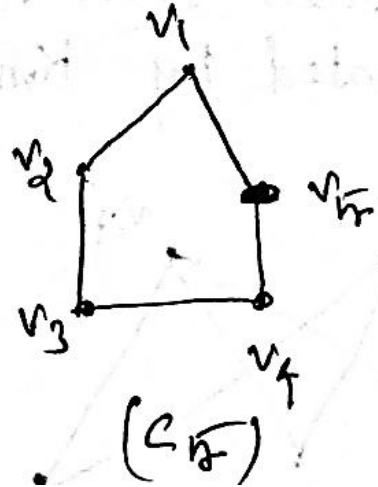
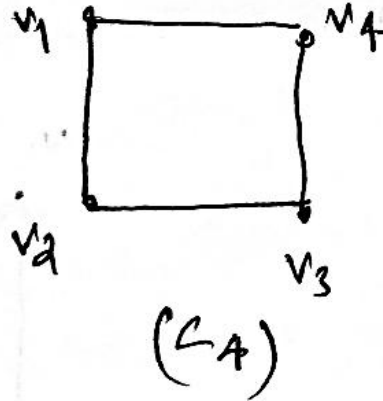
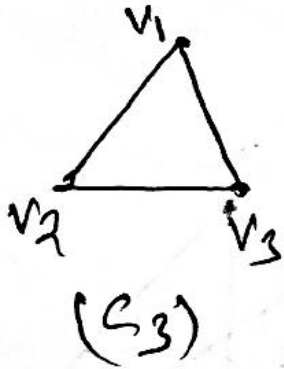
Example
Regular but
not Complete



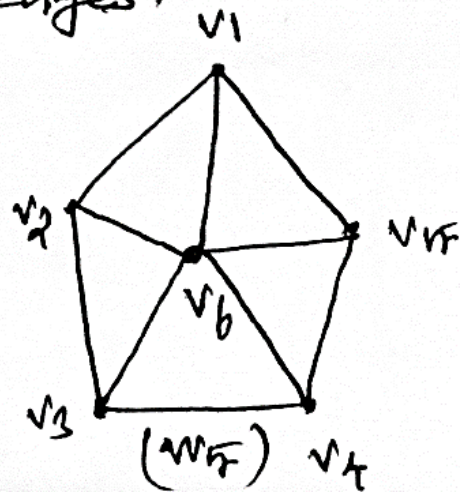
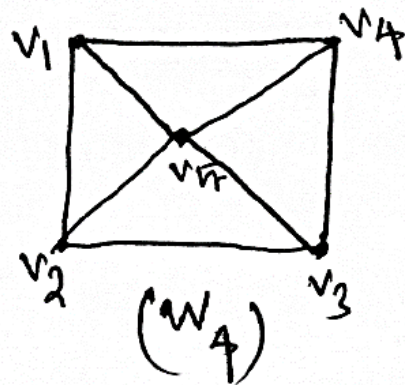
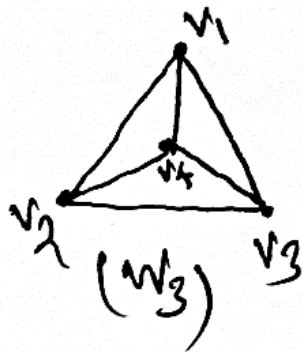
$$\begin{aligned} \deg(v_1) &= \deg(v_2) = \deg(v_3) \\ &= \deg(v_4) = 3 \end{aligned}$$

Cycles

A cycle $C_n, n \geq 3$, consist of n vertices $v_1, v_2, \dots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}$ and $\{v_n, v_1\}$



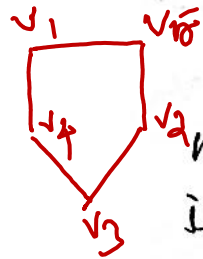
wheels : A wheel W_n is obtained when an additional vertex is added to the cycle C_n , for $n \geq 3$, and this new vertex is connected to each of the n vertices in C_n by the new edges.



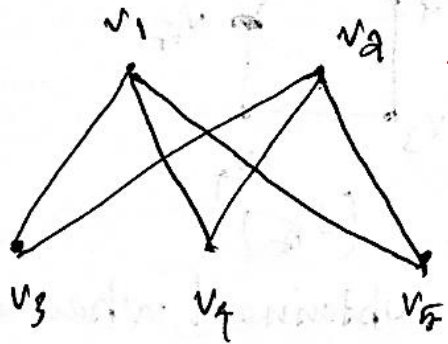
Bipartite Graphs

A graph $G = (V, E)$ is said to be bipartite if the vertex set V can be partitioned into two disjoint subsets V_1 and V_2 such that every edge in E connects a vertex in V_1 and a vertex in V_2 so that no edge in G connects either two vertices in V_1 or two vertices in V_2 . (V_1, V_2) is called bipartition of G .

If further G contains every edge joining the points of V_1 to the points of V_2 then G is called a complete bipartite graph



If V_1 contains m vertices and V_2 contains n vertices then the complete bipartite graph G is denoted by $K_{m,n}$



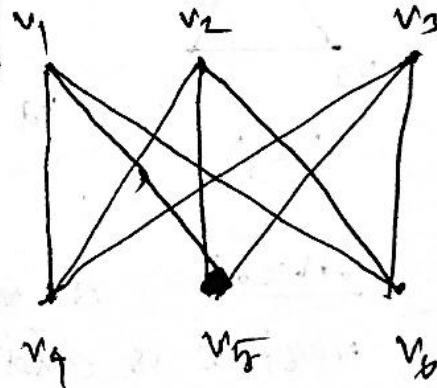
$K_{2,3}$

$$V = \{v_1, v_2, v_3, v_4, v_5\}$$

$$V_1 = \{v_1, v_2\}$$

$$V_2 = \{v_3, v_4, v_5\}$$

$$V = V_1 \cup V_2$$

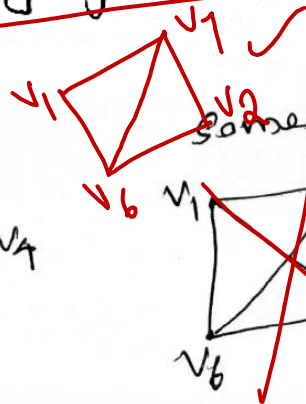
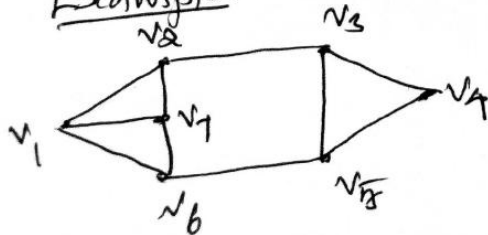


$K_{3,3}$

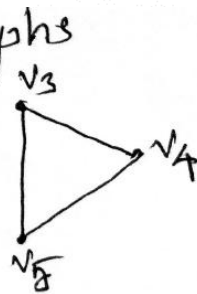
Subgraphs

- ### Subgraphs
- A graph $H = (V_1, E_1)$ is called subgraph of $G = (V, E)$ if $V_1 \subseteq V$ and $E_1 \subseteq E$.
 - A graph $H = (V_1, E_1)$ is called a proper subgraph of $G = (V, E)$ if $V_1 \subset V$ and $E_1 \subset E$.
 - H is called a spanning subgraph of G if $V_1 = V$.
- (Note: A small diagram shows a vertex v_1 with two incident edges, one of which is marked with a checkmark.)*

Example



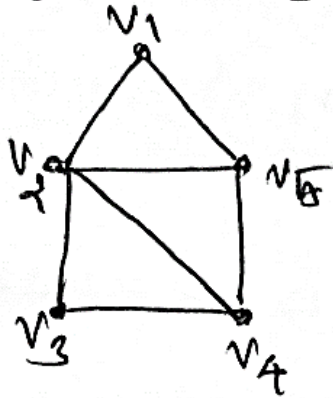
~~some~~ v_2 subgraphs v_3



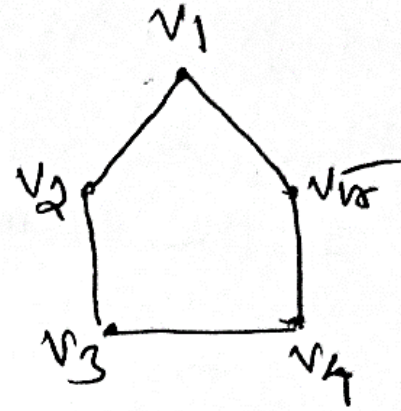
Note

The subgraph which has same vertex set as the original graph is called as Spanning Subgraph.

$V_1 = V$
Condition

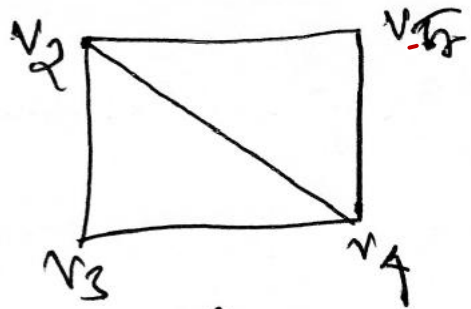


Graph

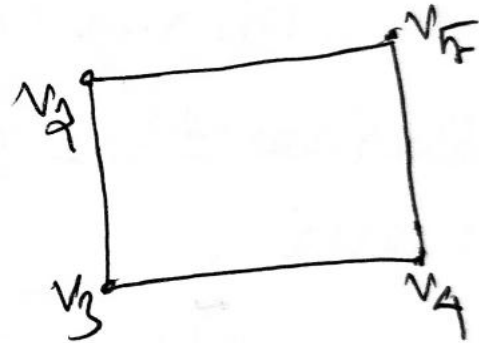


Spanning Subgraph

A subgraph which has all the edges corresponding to those set of vertices is called an induced subgraph



induced subgraph



(not an induced subgraph)

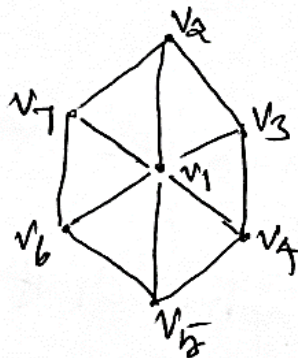
Note : If H is an induced subgraph of G , two points are adjacent in H iff they are adjacent in G .

Vertex deleted subgraph

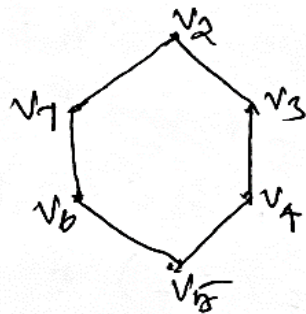
Let $G = (V, E)$ be a graph, let $v_i \in V$.
The subgraph of G obtained by removing the vertex v_i and all the edges incident with v_i is called a vertex deleted subgraph and is denoted by $G - v_i$.

That is, if $G - v_i = (V_i, E_i)$ then $V_i = V - \{v_i\}$
and $E_i = \{e \mid e \in E \text{ and } e \text{ is not incident with } v_i\}$

Graph



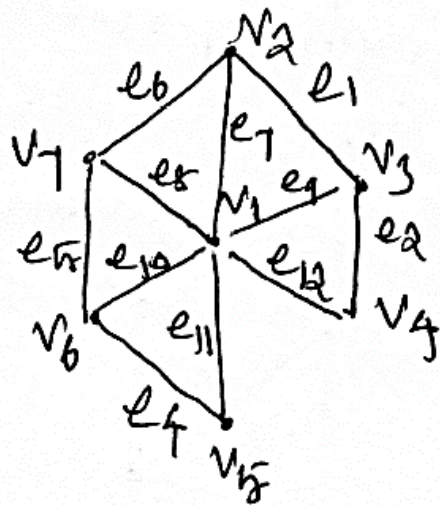
Subgraph $G - v_1$



Edge deleted subgraph

Let $G = (V, E)$ be a graph, let $e_j \in E$. Then $G - e_j = (V, E - \{e_j\})$ is called the subgraph of G obtained by the removal of edge e_j .

subgraph $G - e_j$



Matrix representation of graphs

Adjacency matrix

Let $G = (V, E)$ be a simple graph with n vertices ordered from v_1 to v_n then the adjacency matrix

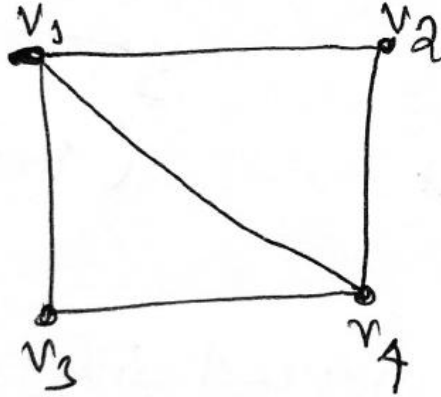
$A = [a_{ij}]_{n \times n}$ of G is an $n \times n$ symmetric matrix ($A = A^T$) defined by the elements

$$a_{ij} = \begin{cases} 1 & \text{when } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise} \end{cases}$$

It is denoted by A_G

Note If the graph G is simple, the degree of the vertex v_i equals the number of 1's in the i th row (or i th column) of A_G

Represent the following graph with an adjacency matrix



Soln.

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Incidence Matrix

Let $G = (V, E)$ be a graph with $v_1, v_2, \dots, v_n$ vertices and $e_1, e_2, \dots, e_m$ edges. Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix

$$M = [m_{ij}]$$

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$

Note

The number of 1's in each row equals to the degree of the corresponding vertex.

Find the incidence matrix of the following graph.

